

2026-1 Computer Graphics
Programming Assignment 2

1 Display multiple polygonal models.

- 1.1 Create **three** 2D polygonal models with **different shapes** and **colors**. [3 pt]
 - ✓ Use more than **four** triangles for each model (e.g., pentagon, house, star, etc.).
 - ✓ There should be **no holes** displayed on the models.
- 1.2 Translate each model on a x, y, and z-axis, respectively. [1 pt]
- 1.3 **Self-rotate** (자전) each model **continuously** against a x, y, and z-axis, respectively. [1 pt]

2 Apply different graphical modes using keyboard inputs.

- 2.1 **r/g/b** keys: change the background color to red/green/blue color. [1 pt]
 - ✓ The background color should be displayed until it is changed.
- 2.2 **w/s** keys: change the polygonal **fill mode** to **wireframe/solid**. [1 pt]
- 2.3 **c** key: turn on/off the **back-face culling** mode. [1 pt]

3 Modify color brightness using keyboard inputs.

- 3.1 **1/2** key: increase/decrease color brightness (+/-10%) by modifying the **pixel shader**. [2 pt]

• **Submission & Cautions**

- ✓ Delete all folders (**Debug, Release, ipch**) and files (***.db, *.exe**) that are **not** necessary for build.
(*폴더 보기 → 표시 → 숨김 항목 → .vs 폴더 → ipch 폴더 및 *.db 파일 삭제)
- ✓ Compress an entire project (**ZIP**, size limit: **10MB**) and submit to **ClassRoom** (***not** accept via E-mail).
- ✓ Use only **Visual Studio 2022** and **DirectX 11**.
- ✓ Build the program in a **x86 (32-bit)** and **Release** mode.
- ✓ Cases of a score of **0**: not using class framework, not building source code, not executing the program, copying source code from others or a previous work, not demonstrating, submissions after the deadline.